

INT4 MAC TRANSFORMER ACCELERATOR

A 4×4 INT4 Tile for Tiny Tapeout on Sky130

A small, weight-stationary INT4 MAC array with an accumulator bank, designed to demonstrate transformer matmul and attention primitives within the area, I/O, and memory constraints of a Tiny Tapeout tile in the Skywater 130 nm open-source PDK.

Block: `int4_mac_accel`

Process: Skywater Sky130 (130 nm)

Platform: Tiny Tapeout (TT)

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Introduction

Transformer models dominate modern machine learning, but the silicon needed to accelerate them is enormous and out of reach for educational tape-outs. The **INT4 MAC Transformer Accelerator** is a deliberate compression of the most essential transformer primitive — small, low-precision matrix multiplication — into a footprint that fits on a single Tiny Tapeout tile in the open-source Skywater 130 nm process.

Rather than attempting a full transformer block, this design isolates the arithmetic kernel that dominates attention and feed-forward layers: a 4×4 array of INT4 $\times$ INT4 multiply-accumulate units feeding a wide accumulator bank. Surrounded by a streaming I/O interface, a small on-chip weight ROM, and an approximate softmax LUT, the block can demonstrate end-to-end attention-style dataflow on tiny dimensions while remaining honest about what fits in roughly 25,000–35,000 standard cells.

This whitepaper presents the architecture, microarchitecture, and physical design intent of the block, in the style of a production silicon specification. It is intended as a source of truth for RTL design, verification, physical implementation, and post-silicon bring-up, and as a teaching reference for accelerator design at Tiny Tapeout scale.

DESIGN PHILOSOPHY

Build the smallest honest transformer-shaped accelerator that fits — no marketing asterisks, no off-chip DRAM assumptions. Constraints define the architecture.

Key Features

The accelerator is built around five tightly-coupled features, each chosen to maximize useful computation per gate within Tiny Tapeout's roughly 1 mm² area envelope and ~16-pin I/O budget.

4×4 INT4 MAC Array

16 weight-stationary processing elements compute 16 INT4 multiplies per cycle, accumulating into INT20 partial sums. The fundamental matmul kernel of attention and FFN layers, sized to fit.

32-Entry INT20 Accumulator Bank

Dual-banked accumulators allow tiles larger than the array to be computed across multiple passes, enabling matmuls of practical sequence lengths without overflow.

Streaming I/O Interface

An 8-bit packetized streaming protocol over the TT pin budget delivers activations and accepts results with explicit flow control. No DRAM, no wide buses.

On-Chip Weight ROM

Up to 256 INT4 weights baked as synthesized constants. Constant-folding collapses multiplier area dramatically, enabling a fully self-contained fixed-network demonstration.

Approximate Softmax / Activation

16-entry piecewise LUT for exp/softmax approximation and a small requantization unit close the inference loop entirely on-chip.

Tiny Tapeout Native

Designed against TT pin maps, OpenLane synthesis, and a 25–50 MHz target clock. Reproducible from public PDKs and tools end-to-end.

Architecture In-Depth

Top-Level Block Diagram

The accelerator is organized as a control plane (FSM and I/O), a data plane (MAC array and accumulator bank), and a small post-processing path (requantization and activation LUT). The figure below shows the top-level structure and the major interfaces.

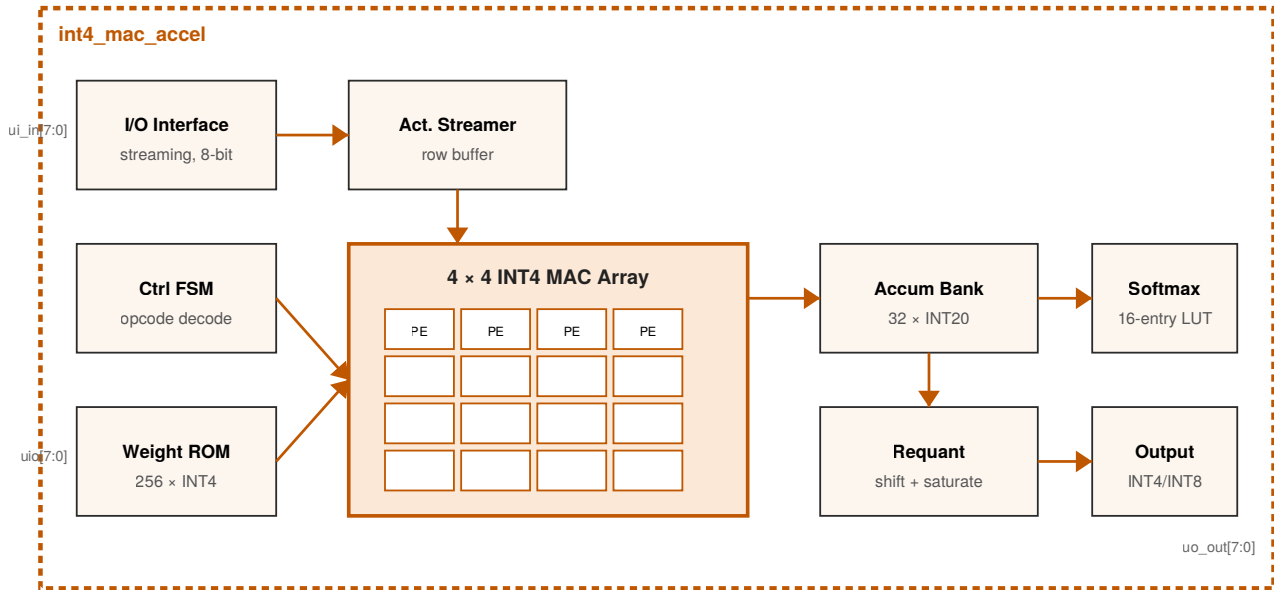


Figure 1. `int4_mac_accel` top-level block diagram. Submodule layout maps 1:1 to the top-level RTL module.

MAC Array Microarchitecture

Each of the 16 processing elements (PEs) holds one INT4 weight register, performs one signed INT4×INT4 multiply per cycle, and accumulates into a local INT20 partial sum. The array is **weight-stationary**: weights are loaded once into the PEs and held for many activation passes, mirroring the dataflow of K and V projections across a sequence of input tokens.

Parameter	Value	Notes
Array shape	4 rows × 4 cols	16 PEs total
PE multiplier	INT4 × INT4 signed	~300–500 gates per PE
PE accumulator	INT20	Headroom for ~64 partial sums without overflow
Throughput	16 MACs / cycle	32 ops/cycle (mul + add)
Peak rate	~0.8 GOPS @ 25 MHz	~1.6 GOPS at 50 MHz
Estimated gates	~10,000	Excluding accumulator bank

Accumulator Bank

A 32-entry bank of INT20 accumulators sits downstream of the array, allowing output tiles larger than 4×4 to be computed by accumulating across multiple array passes. This is the single most important feature for supporting useful matmul sizes: without it, the array can only produce one 4×4 output tile per weight load.

Requantization & Activation Path

After accumulation, results pass through a configurable shift-and-saturate requantization unit that converts INT20 partial sums back to INT4 or INT8 for the next layer. An optional 16-entry LUT approximates softmax (exp + reciprocal-sum) or GELU/SiLU activations, depending on synthesis-time configuration.

Performance & Area

The following targets reflect a Sky130 Tiny Tapeout 8× tile budget, synthesized through OpenLane at a 25 MHz target clock. They are first-order estimates; final numbers will be reported post-place-and-route.

Metric	Target	Comment
Process	Skywater Sky130	Open-source PDK, 130 nm
Tile size	8× TT tile (~1280 × 800 μm)	~1.0 mm ² usable
Target clock	25 MHz	50 MHz stretch
Standard cells (est.)	25,000 – 35,000	Synthesis estimate
Flop count (est.)	~2,500	Includes accumulators and pipeline registers
SRAM macros	0 – 1	Optional 256 B activation buffer (OpenRAM)
Pin budget	8 in + 8 out + 8 bidir	Standard TT pinout
Peak throughput	0.8 GOPS @ 25 MHz	16 INT4 MACs/cycle
Sustained throughput	~0.4 GOPS	I/O-bound on streaming activations
Unloaded latency	~6 cycles	Input → first INT4 output
Power (typical)	~5 mW	Estimated, dominated by array switching

WHY I/O IS THE REAL BOTTLENECK

With only 8 dedicated input pins, sustained activation streaming at 25 MHz delivers 25 MB/s — sufficient for one 4-element INT4 vector every 2 cycles. Peak compute far exceeds peak I/O, which is why on-chip weight ROM and accumulator reuse matter more than adding more MACs.

Microarchitecture Specification Skeleton

The remainder of this document is a working microarchitecture specification skeleton organized per the Anthropic-style template: change log, parameters, interfaces, clocks/resets, performance, physical design, and per-submodule implementation detail. Sections are stubbed with table shapes and placeholder rows, ready to be filled in as the design firms up.

Change Log

Date	Version	Description	Authors	Reviewers
2026-05-04	v0.1	Initial draft	<i>TBD</i>	<i>TBD</i>

Parameters & Typedefs

Parameter	Type	Default	Description
ARRAY_ROWS	int	4	MAC array rows
ARRAY_COLS	int	4	MAC array columns
DATA_W	int	4	Activation/weight bit width
ACC_W	int	20	Accumulator bit width
ACC_DEPTH	int	32	Accumulator bank entries
ROM_ENTRIES	int	256	On-chip weight ROM depth

Top-Level Interfaces

Interface	Direction	Width	Description
<code>ui_in[7:0]</code>	input	8	Streaming activation / opcode input
<code>uo_out[7:0]</code>	output	8	Result / status output
<code>uio_in[7:0]</code>	input	8	Bidirectional control (input mode)
<code>uio_out[7:0]</code>	output	8	Bidirectional control (output mode)
<code>uio_oe[7:0]</code>	output	8	Bidirectional output enable
<code>clk</code>	input	1	Primary clock (25–50 MHz)
<code>rst_n</code>	input	1	Active-low synchronous reset
<code>ena</code>	input	1	Tile enable (TT framework)

Clocks & Resets

Signal	Kind	Frequency / Polarity	Description
<code>clk</code>	Primary clock	25 MHz nominal, 50 MHz stretch	Single clock domain across all submodules
<code>rst_n</code>	Synchronous, active-low	Synchronous to <code>clk</code>	Resets all flops; weight ROM is non-resettable

Performance Requirements

First-Order: Steady-State Throughput

Sustained throughput is bounded by activation streaming bandwidth. With one INT4 activation arriving every cycle on `ui_in` (4 bits used of 8), the array consumes 4 activations per cycle once primed, achieving 16 MACs/cycle during compute bursts.

First-Order: Unloaded Latency

Input activation to first output: **~6 cycles** (1 stream-in, 2 array pipeline, 1 accumulator write, 1 requant, 1 stream-out).

Second-Order: Loaded Latency

With back-to-back tile computations, loaded latency for a 16×16 INT4 matmul is approximately 80–100 cycles, dominated by the four 4×4 sub-tile passes plus accumulator bank writes.

Physical Design

Item	Plan
Tile size	$8 \times$ TT tile
Floorplan	I/O along left edge; MAC array centered; accumulator bank to the right
SRAM	Optional $1 \times$ OpenRAM 256B 8-bit activation buffer (TBD)
Weight storage	Synthesized constants (no SRAM/ROM macro)
Critical path	Multiplier $\rightarrow$ adder tree $\rightarrow$ accumulator write

CSRs, Debug, Reliability

A minimal CSR set is exposed via the streaming opcode interface: `STATUS`, `CTRL`, `ACC_READ`, and `MODE`. Debug is provided via observable accumulator readback. No ECC; reliability targets are functional correctness over a single-die educational tape-out.

References

- Tiny Tapeout: <https://tinytapeout.com>
- Skywater Sky130 PDK: <https://github.com/google/skywater-pdk>
- OpenLane: <https://github.com/The-OpenROAD-Project/OpenLane>
- OpenRAM: <https://openram.org>
- NVIDIA Ampere GA102 GPU Architecture Whitepaper (style reference)
- Anthropic Microarchitecture Spec Template (organizational reference)

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